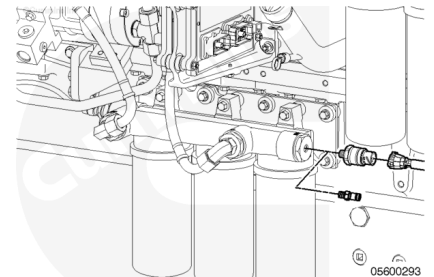


Trainings ▾

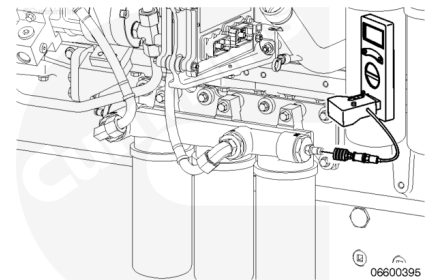
Initial Check

Note : Shown is the Stage 2 filter head for the QSK38. Although different in appearance, the procedure remains the same.

Remove the pressure sensor and replace it with a Compuchek™ fitting.

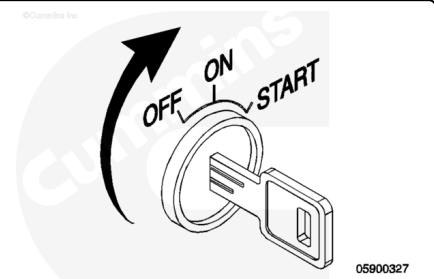


Connect the pressure adapter, Part Number 3164491, or equivalent, and the digital multimeter, Part Number 3164488, 3164489, or equivalent, to the Compuchek™ fitting.



The fuel lift pump will operate for 1000 seconds when the keyswitch is in the ON position. The fuel lift pump will also operate while the engine is cranking, until the fuel rail pressure reaches 200 bar [2900 psi].

At the end of 1000 seconds of operation, the fuel lift pump **must** produce pressure. If the lift pump does **not**, then the lift pump **must** be replaced.



Minimum Fuel Lift Pump Pressure

bar	psi
3	43.5

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery cables. See equipment manufacturer service information.
- Close the fuel supply valve.

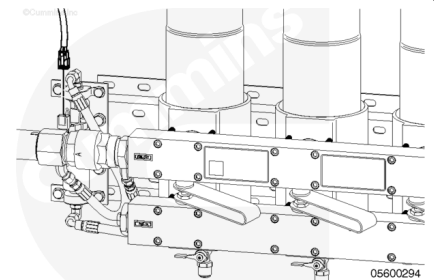


Remove

Off-Engine

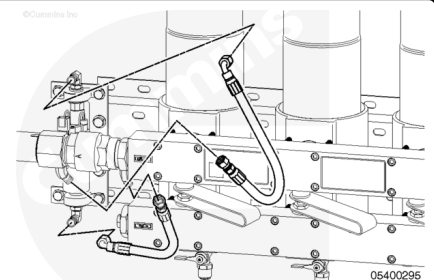
Note : Shown in this procedure are the marine application Stage 1 filters. Although different in appearance, the lift pump procedure remains the same.

Disconnect the electrical connection to the fuel lift pump.



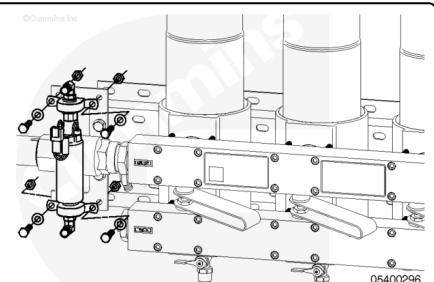
⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting to performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke, or allow sparks or flames (such as pilot lights, arcing equipment, or welding equipment) in the work area.

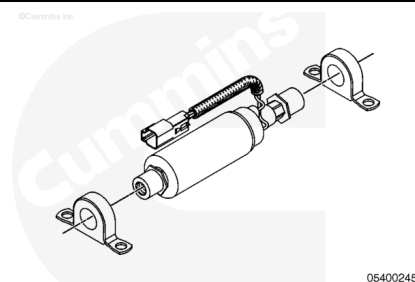


Disconnect the supply and discharge flexible hoses from the lift pump.

Remove the four capscrews securing the fuel lift pump to the fuel filter mounting bracket.



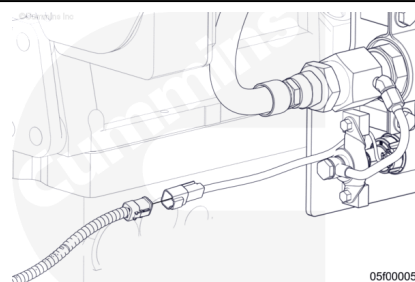
Remove the vibration isolators from each end of the fuel lift pump.



05400245

On-Engine

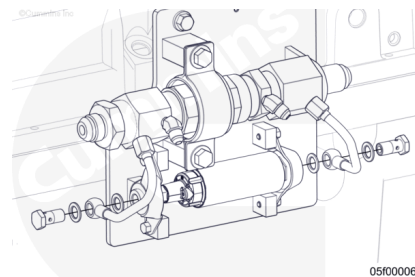
Disconnect the electrical connection to the fuel lift pump.



05R00005

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting to performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke, or allow sparks or flames (such as pilot lights, arcing equipment, or welding equipment) in the work area.

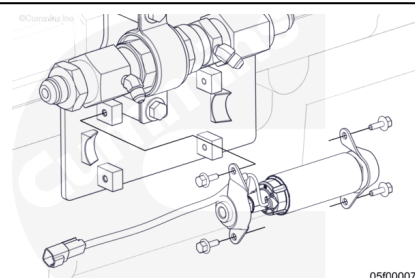


05R00006

Remove the banjo capscrews and sealing washers and loosen the tube fittings securing the supply and discharge fuel lines from the lift pump.

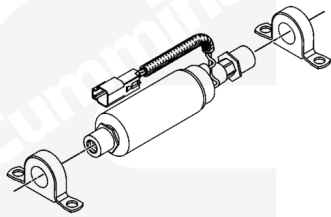
Remove the fuel lines.

Remove the four capscrews securing the fuel lift pump to the mounting bracket.



05R00007

Remove the vibration isolators from each end of the fuel lift pump.



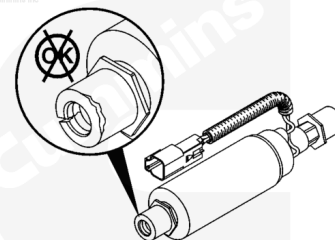
05400245

Inspect for Reuse

Inspect the fuel lift pump for any signs of damage.

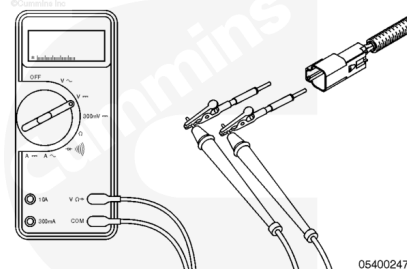
Replace the fuel lift pump, if damaged.

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05400246

Use a digital multimeter to measure the resistance across the pins in the electrical connector. Reference the table for resistance value. If **not** within specification, replace the fuel lift pump.



05400247

Fuel Lift Pump Resistance

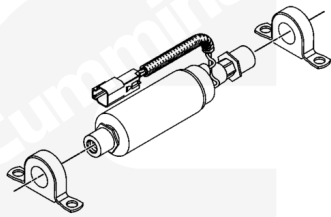
Minimum	Maximum
1 ohm	3 ohm

Use a digital multimeter to check for a short between each pin and the fuel lift pump body. Replace the fuel lift pump, if shorted.

Install

Off-Engine

Install vibration isolators to each end of the fuel lift pump.

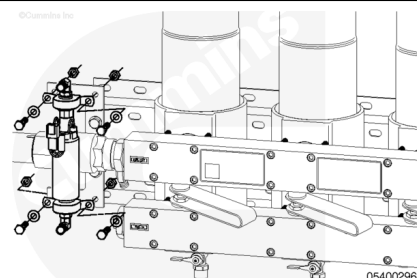


05400245

Install the fuel lift pump and four capscrews to the fuel filter mounting bracket.

Torque Value: 15 n•m [133 in-lb]

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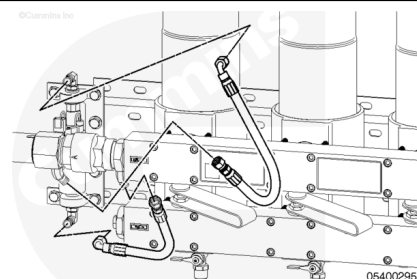


05400296

Connect the supply and discharge flexible hoses to the lift pump.

Torque Value: 54 n•m [40 ft-lb]

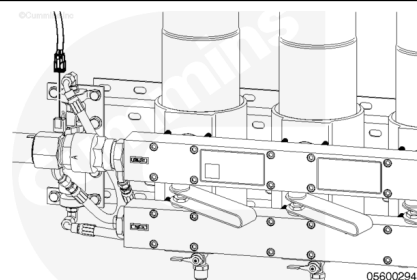
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05400295

Connect the electrical connection to the fuel lift pump.

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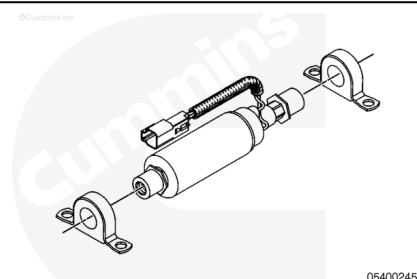


05600294

On-Engine

Install vibration isolators to each end of the fuel lift pump.

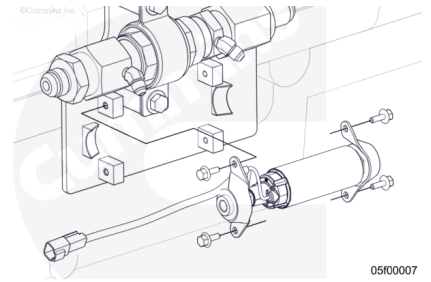
©Cummins Inc.



05400245

Install the fuel lift pump and four capscrews to the fuel filter mounting bracket.

Torque Value: 15 n•m [133 in-lb]

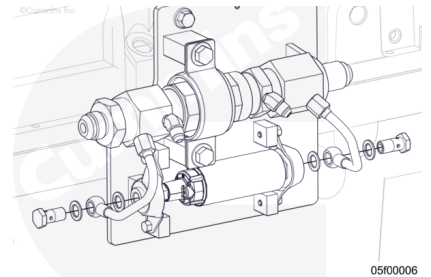


Install the banjo capscrews and new sealing washers securing the fuel supply and discharge lines to the lift pump.

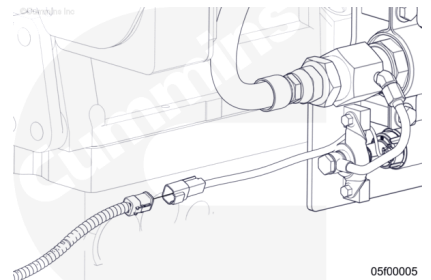
Torque Value: 60 n•m [44 ft-lb]

Rotate tubes until tube fittings are aligned with 45 degree fittings on check valve. Tighten fittings.

Torque Value: 20 n•m [177 in-lb]



Connect the electrical connection to the fuel lift pump.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the battery cables. See equipment manufacturer service information.
- Open the fuel supply valve
- Turn the keyswitch to the ON position.
- The fuel lift pump will operate for 1000 seconds.
- Check the fuel lift pump connections for leaks.

